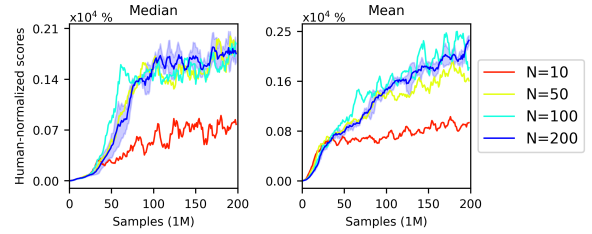


(a) Different kernel bandwidths (at fixed $N = 200$).



(b) Different values of N (at fixed $h = \text{mix}$).

Figure 3: The sensitivity of (the human-normalized scores of) MMDQN in the 6 tuning games with respect to: (a) the kernel choices (Gaussian kernels with different bandwidths h and unrectified kernels), and (b) the number of particles N . Here $h = \text{mix}$ indicates the mixture of bandwidth values in $\{1, 2, \dots, 10\}$. All curves are smoothed over 5 consecutive iterations. 95% C.I. for the $h = \text{mix}$, $N = 200$, and unrectified kernel curves (3 seeds) and 1 seed for the other curves.

Baselines. We categorize the baselines into two groups. The first group contains *comparable* methods: DQN, PRIOR., C51, and QR-DQN-1, where DQN (Mnih et al. 2015) and PRIOR. (prioritized experience replay (Schaul et al. 2016)) are classic baselines. The second group includes *reference* methods: RAINBOW (Hessel et al. 2018), IQN, and FQF, which contain algorithmic/modeling improvements orthogonal to MMDQN: RAINBOW combines C51 with prioritized replay and n -step update while IQN and FQF contain modeling improvements as described in the related work section. Since in this work we used the same architecture as QR-DQN and C51 for MMDQN, we directly compare MMDQN with the first group while including the second group for reference.

Hyperparameter setting. For fair comparison with QR-DQN, we used the same hyperparameters: $N = 200$, Adam optimizer (Kingma and Ba 2015) with $lr = 0.00005$, $\epsilon_{ADAM} = 0.01/32$. We used ϵ -greedy policy with ϵ being decayed at the same rate as in DQN but to a lower value $\epsilon = 0.01$ as commonly used by the distributional RL methods. We used a target network to compute the distributional Bellman target as with DQN. Our implementation is based on OpenAI Gym (Brockman et al. 2016) and the Dopamine framework (Castro et al. 2018).

Kernel selection. We used Gaussian kernels $k_h(x, y) = \exp(-(x - y)^2/h)$ where $h > 0$. The kernel bandwidth h is crucial to the statistical quality of MMD: overestimated bandwidth results in a flat kernel while underestimated one makes the decision boundary highly irregular. We utilize the kernel mixture trick in (Li, Swersky, and Zemel 2015) which is a mixture of K kernels covering a range of bandwidths $k(x, y) = \sum_{i=1}^K k_{h_i}(x, y)$. The Gaussian kernel with a bandwidth mixture yields much a better performance than that with individual bandwidth and unrectified kernels in 6 tuning games: Breakout, Assault, Asterix, MsPacman, Qbert, and BeamRider (see Figure 3 (a)). Figure 3 (b) shows the sensitivity of MMDQN in terms of the number of particles N in the 6 tuning games where too small N adversely affects the performance.

The main empirical result is provided in Table 1 where we compute the mean and median of *best* human normalized scores across 55 Atari games in the 30 no-op eval-

uation setting (the full raw scores for each game are provided in Appendix C). The table shows that MMDQN significantly outperforms the comparable methods in the first group (DQN, PRIOR., C51 and QR-DQN-1) in all metrics though it shares the same network architecture with C51 and QR-DQN-1. Although we did not include any orthogonal algorithmic/modelling improvements from the reference methods in the second group to MMDQN, MMDQN still performs comparably with these methods and even achieve a state-of-the-art mean human-normalized score. In Figure 4 we also provide the percentage improvement per-game of MMDQN over QR-DQN-1 where MMDQN offers significant gains over QR-DQN-1 in a large array of games.

	Mean	Median	>Human	>DQN
DQN	221%	79%	24	0
PRIOR.	580%	124%	39	48
C51	701%	178%	40	50
QR-DQN-1	902%	193%	41	54
RAINBOW	1213%	227%	42	52
IQN	1112%	218%	39	54
FQF	1426%	272%	44	54
MMDQN	1969%	213%	41	55

Table 1: Mean and median of *best* human-normalized scores across 55 Atari 2600 games. The results for MMDQN are averaged over 3 seeds and the reference results are from (Yang et al. 2019).

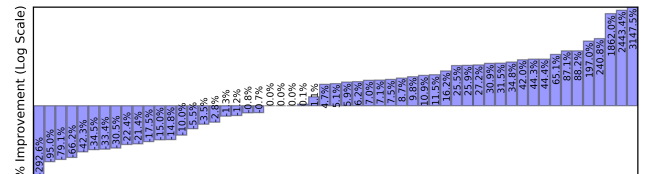


Figure 4: Percentage improvement per-game of MMDQN over QR-DQN-1.

Conclusion and Discussion

We have introduced a novel approach for distributional RL that eschews the predefined statistic principle used in the prior distributional RL. Our method deterministically evolves the (pseudo-)samples of a return distribution to approximately match moments of the resulting approximate distribution with those of the return distribution. We have also provided theoretical understanding of distributional RL within this framework. Our experimental results show that MMDQN, a combination of our approach with DQN-like architecture, achieves significant improvement in the Atari benchmark.

In what follow, we further discuss our results and the future work.

Deterministic sampling in Theorem 3. Theorem 3 is concerned with the convergence of the optimal deterministic particles uniquely arisen in our problem setting where we deterministically evolve a set of particles to approximate a distribution in MMD. In other words, Theorem 3 does not fully analyze Algorithm 1 but addresses one relevant yet important aspect: if a set of deterministic particles are evolved to simulate a distribution in MMD, how good is the approximation. This is different from the conventional setting in MMD which are often concerned with the convergence of empirical MMD derived from i.i.d. samples of each component distribution (Gretton et al. 2012) (though we leverage similar proof techniques). We also remark that in Theorem 3, we assume the attainability of the infimum but do not specify a practical algorithm to solve this infimum. While in practice, we use neural networks to represent the deterministic particles and use SGD to solve this optimization problem (as in MMDQN), the related literature of kernel herding can in fact provide a different approach with an improved analysis. Herding (Welling 2009) is a method that generates pseudo-samples (i.e., deterministic samples) from a distribution such that nonlinear moments of the sample set closely match those of the target distribution. A greedy selection of pseudo-samples can achieve a convergence rate of $O(1/n)$ (Chen, Welling, and Smola 2010). Different from greedy herding, MMDQN collectively find the set of pseudo-samples using SGD at each learning step. This collective herding by SGD is more effective in the distributional RL context than greedy herding as in distributional RL we need to perform herding for multiple distributions which themselves also evolve over learning steps.

Predefined statistics vs. unrestricted statistics. We further clarify that a specified functional form in Section “Predefined statistic principle” section means that the parametric form ζ , which possibly depends on some real parameters, is fully specified, e.g., quantile values $F_{\eta}^{-1}(\tau)$ (which depends on quantile level τ) are predefined statistics. In this sense, FQF/IQN still have predefined statistics as they use quantile values to approximate a distribution. We emphasize that the flexibility of τ in FQF/IQN does not solve the problem of predefined statistics but rather only gives a finer-grained approximation by using more τ which in turn increases the budget of N statistics. We

remark that a flexible increase of N (which can give a finer approximation in both predefined and unrestricted statistics) is not our focus in this paper; thus, we find it useful to keep the same fixed budget of N statistics when comparing predefined and unrestricted statistics. We also remark that a general empirical sample can be considered an unrestricted statistic as it does not subscribe to any predefined parametric functional form. As a concrete example, let us consider the task of approximating a distribution with only one (learnable) statistic: One approach targets the median, and the other uses an unrestricted statistic. Eventually, the former approach should converge its statistic to the median as it subscribes to this predefined functional form while the latter approach can evolve its statistic into any sample equally (not necessarily the median) as long as the sample can simulate the distribution in a certain sense (in our case, to match the moments of the empirical distribution with those of the target distribution).

Automatic kernel selection for MMDQN. The kernel used in MMDQN plays a crucial role in achieving a good empirical performance and selecting the same kernel to perform well in all the games is a highly non-trivial task. Our current work uses a relatively simple but effective heuristics which uses a mixture of Gaussian kernels with different bandwidths. We speculate that a systematic way of selecting a kernel can even boost the empirical performance of MMDQN further. A promising direction is that instead of relying on a predefined kernel, we can train an adversarial kernel (Sriperumbudur et al. 2009; Li et al. 2017) to provide a stronger signal about a discrepancy between two underlying distributions; that is, $\min_{\theta \in \Theta} \max_{k \in \mathcal{K}} \text{MMD}(Z_{\theta}(x, a), [\mathcal{T}Z_{\theta}](x, a); k), \forall (x, a)$ where $\mathcal{K}$ is a set of kernels.

Modeling improvements for MMDQN. We focus our current work only on the statistical aspect of distributional RL and deliberately keep all the other design choices similar to the basic QR-DQN (e.g., we did not employ any modeling improvements and uncertainty-based exploration). As our framework does not require the likelihood but only (pseudo-)samples from the return distribution, it is natural to build an implicit generative model (as in IQN) for the return distribution in MMDQN where we transform via a deterministic parametric function the samples from a base distribution, e.g., a simple Gaussian distribution, to the samples of the return distribution. The weights of the empirical distribution can be made learnable by a proposal network as in FQF.

An open question about the sufficient condition for contraction. In this work, we prove that the distributional Bellman operator is not a contraction in MMD with Gaussian kernels using the scale-insensitivity of Gaussian kernels. On the other hands, we show that the distributional Bellman operator is a contraction in MMD with shift-invariant and scale-sensitive kernels. This suggests a question of whether the scale sensitivity is a necessary condition for the contraction under MMD. Another direc-

tion is an understanding of a precise notion and the role of approximate contraction in practical setting as here the distributional Bellman operator is not a contraction in MMD with Gaussian kernels but the Gaussian kernels still give a favorable empirical performance in the Atari games as compared to the other kernels.

Robust off-policy estimation in distributional RL. Another potential direction from the current work is to estimate the return distributions merely from offline data generated by some behaviour policies. Since the estimation is constructed from finite offline data, robustness is a key to avoid a spurious estimation (Nguyen et al. 2020).

Acknowledgement

This research was partially funded by the Australian Government through the Australian Research Council (ARC). Prof. Venkatesh is the recipient of an ARC Australian Laureate Fellowship (FL170100006). We would also like to thank our anonymous reviewers (from NeurIPS’20 and AAAI’21) for the constructive feedback, Will Dabney (DeepMind) for the raw result data of QR-DQN-1 and the useful comments for our initial draft, and Mengyan Zhang (Australian National University) for the interesting discussion about order statistics.

References

- Banach, S. 1922. Sur Les Operations Dans Les Ensembles Abstraits et Leur Application Aux Equations Integrales.
- Bellemare, M. G.; Dabney, W.; and Munos, R. 2017. A Distributional Perspective on Reinforcement Learning. In *ICML*, volume 70 of *Proceedings of Machine Learning Research*, 449–458. PMLR.
- Bellemare, M. G.; Naddaf, Y.; Veness, J.; and Bowling, M. 2013. The Arcade Learning Environment: An Evaluation Platform for General Agents. *Journal of Artificial Intelligence Research* 47: 253–279. ISSN 1076-9757. doi: 10.1613/jair.3912.
- Bellemare, M. G.; Roux, N. L.; Castro, P. S.; and Moitra, S. 2019. Distributional reinforcement learning with linear function approximation.
- Bellman, R. 1957. *Dynamic Programming*. Princeton, NJ, USA: Princeton University Press, 1 edition.
- Brockman, G.; Cheung, V.; Pettersson, L.; Schneider, J.; Schulman, J.; Tang, J.; and Zaremba, W. 2016. OpenAI Gym.
- Castro, P. S.; Moitra, S.; Gelada, C.; Kumar, S.; and Bellemare, M. G. 2018. Dopamine: A Research Framework for Deep Reinforcement Learning. *CoRR* abs/1812.06110.
- Chen, W. Y.; Mackey, L. W.; Gorham, J.; Briol, F.; and Oates, C. J. 2018. Stein Points. In *ICML*, volume 80 of *Proceedings of Machine Learning Research*, 843–852. PMLR.
- Chen, Y.; Welling, M.; and Smola, A. J. 2010. Super-Samples from Kernel Herding. In *UAI*, 109–116. AUAI Press.
- Chung, K.-J.; and Sobel, M. J. 1987. Discounted MDP’s: distribution functions and exponential utility maximization.
- Dabney, W.; Ostrovski, G.; Silver, D.; and Munos, R. 2018a. Implicit Quantile Networks for Distributional Reinforcement Learning. In *ICML*, volume 80 of *Proceedings of Machine Learning Research*, 1104–1113. PMLR.
- Dabney, W.; Rowland, M.; Bellemare, M. G.; and Munos, R. 2018b. Distributional Reinforcement Learning With Quantile Regression. In *AAAI*, 2892–2901. AAAI Press.
- Fournier, N.; and Guillin, A. 2015. On the rate of convergence in Wasserstein distance of the empirical measure. *Probability Theory and Related Fields* 162(3–4): 707–738.
- Fukumizu, K.; Gretton, A.; Sun, X.; and Schölkopf, B. 2007. Kernel Measures of Conditional Dependence. In *NIPS*, 489–496. Curran Associates, Inc.
- Gretton, A.; Borgwardt, K. M.; Rasch, M. J.; Schölkopf, B.; and Smola, A. J. 2012. A Kernel Two-Sample Test. *J. Mach. Learn. Res.* 13: 723–773.
- Hessel, M.; Modayil, J.; van Hasselt, H.; Schaul, T.; Ostrovski, G.; Dabney, W.; Horgan, D.; Piot, B.; Azar, M. G.; and Silver, D. 2018. Rainbow: Combining Improvements in Deep Reinforcement Learning. In *AAAI*, 3215–3222. AAAI Press.
- Kingma, D. P.; and Ba, J. 2015. Adam: A Method for Stochastic Optimization. In *ICLR (Poster)*.
- Li, C.; Chang, W.; Cheng, Y.; Yang, Y.; and Póczos, B. 2017. MMD GAN: Towards Deeper Understanding of Moment Matching Network. In *NIPS*, 2203–2213.
- Li, Y.; Swersky, K.; and Zemel, R. S. 2015. Generative Moment Matching Networks. In *ICML*, volume 37 of *JMLR Workshop and Conference Proceedings*, 1718–1727. JMLR.org.
- Liu, Q.; and Wang, D. 2016. Stein Variational Gradient Descent: A General Purpose Bayesian Inference Algorithm. In *NIPS*, 2370–2378.
- Lyle, C.; Bellemare, M. G.; and Castro, P. S. 2019. A Comparative Analysis of Expected and Distributional Reinforcement Learning. In *AAAI*, 4504–4511. AAAI Press.
- Mnih, V.; Kavukcuoglu, K.; Silver, D.; Rusu, A. A.; Veness, J.; Bellemare, M. G.; Graves, A.; Riedmiller, M.; Fidjeland, A. K.; Ostrovski, G.; et al. 2015. Human-level control through deep reinforcement learning. *Nature* 518(7540): 529–533.
- Morimura, T.; Sugiyama, M.; Kashima, H.; Hachiya, H.; and Tanaka, T. 2010a. Nonparametric Return Distribution Approximation for Reinforcement Learning. In *ICML*, 799–806. Omnipress.
- Morimura, T.; Sugiyama, M.; Kashima, H.; Hachiya, H.; and Tanaka, T. 2010b. Parametric Return Density Estimation for Reinforcement Learning. In *UAI*, 368–375. AUAI Press.
- Nguyen, T. T.; Gupta, S.; Ha, H.; Rana, S.; and Venkatesh, S. 2020. Distributionally Robust Bayesian Quadrature Optimization. In Chiappa, S.; and Calandra, R., eds., *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings*